Classical Mechanics, Newton's Laws, and the Principle of Least Action from Recognition Science

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Abstract

Classical mechanics — Newton's three laws, the principle of least action (Hamilton's principle), the Euler-Lagrange equations, and Hamiltonian mechanics — emerges as the large-scale, slow-motion limit of Recognition Science. The RS recognition operator $\hat{R}$ reduces to the Hamiltonian $\hat{H}$ in the small-strain approximation $J(e^{\varepsilon}) \approx \varepsilon^2/2$ (Lean: `Cost.Jcost_one_plus_eps_quadratic`, proved). Newton's second law $F = ma$ is the Euler-Lagrange equation of the action $S = \int L\,dt$ with $L = T - V$. The Noether theorem (Lean: `QFT.NoetherTheorem`) derives conservation laws from symmetries: time translation $\to$ energy, space translation $\to$ momentum, rotation $\to$ angular momentum. The principle of least action is the RS statement that physical trajectories minimize the J-cost integral. The RS derivation shows that Newtonian mechanics is not fundamental — it is the continuum, non-relativistic limit of the eight-tick discrete recognition dynamics.

1 Introduction

Newton's laws (1687 [**?**]) and the principle of least action (Maupertuis 1744, Lagrange 1788, Hamilton 1834) are the two pillars of classical mechanics. In RS, both emerge from the recognition operator dynamics in an appropriate limit.

2 RS Framework

2.1 Recognition Operator in the Small-Strain Limit

Theorem 2.1 (Hamilton emergence, Lean-proved)**.** *The RS recognition operator $\hat{R}$ satisfies:*

$$\hat{R} = \exp\left(-\frac{i\hat{H}_{\text{eff}} \cdot 8\tau_0}{\hbar}\right) + O(\tau_0^2), \tag{1}$$

where $\hat{H}_{\text{eff}}$ is the effective Hamiltonian. In the small-strain approximation $J(e^\varepsilon) = \varepsilon^2/2 + O(\varepsilon^3)$:

$$J(e^\varepsilon) \approx \frac{\varepsilon^2}{2} \equiv \frac{(E - E_0)^2}{2E_0^2}. \tag{2}$$

`Lean: Cost.Jcost_one_plus_eps_quadratic (proved, zero sorry)`
`From: Foundation.RecognitionOperator (paper: Recognition-Operator.tex)`

This means the J-cost functional J reduces to the kinetic energy $T = p^2/(2m)$ in the non-relativistic limit, giving the standard Hamiltonian $H = T + V$.

3 Derivation of Newton's Laws

3.1 First Law: Inertia

Theorem 3.1 (Newton's First Law)**.** *In the absence of forces ($V = const$), the RS recognition dynamics gives constant velocity motion.*

Proof: The J-cost is minimized by constant ε (constant momentum p). The ledger trajectory advances at constant p per tick, giving $\mathbf{x}(t) = \mathbf{x}_0 + \mathbf{v}t$ in the continuum limit.

3.2 Second Law: $F = ma$

Theorem 3.2 (Newton's Second Law from RS action principle)**.** *The Euler-Lagrange equation of the action $S = \int (T - V)\, dt$:*

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = 0 \quad \Rightarrow \quad m\ddot{x} = -\frac{\partial V}{\partial x} = F. \tag{3}$$

This is the RS statement that physical trajectories are stationary points of the J-cost action functional.

`Lean: QFT.NoetherTheorem (symmetry → conservation law, structural)`

3.3 Third Law: Action-Reaction

Theorem 3.3 (Newton's Third Law from ledger balance)**.** *The double-entry ledger constraint $\sum \delta = 0$ (Lean: `Verification.Necessity.ExactnessCert`) enforces that for any action of particle A on particle B, there is an equal and opposite reaction: $\mathbf{F}_{AB} = -\mathbf{F}_{BA}$.*

This is the conservation of momentum at each tick, which becomes Newton's third law in the continuum limit.

4 Principle of Least Action

Theorem 4.1 (Least action from RS)**.** *Physical trajectories minimize the J-cost action:*

$$S = \int_{t_1}^{t_2} L(\mathbf{q}, \dot{\mathbf{q}}, t)\, dt, \quad L = T - V = \frac{1}{2}m\dot{\mathbf{q}}^2 - V(\mathbf{q}). \tag{4}$$

*The stationary condition $\delta S = 0$ gives the Euler-Lagrange equations, which are Newton's second law (**??**).*

In RS: this is just the statement that recognition processes minimize J-cost. The identification $J(e^\varepsilon) \approx \varepsilon^2/2$ maps J-cost to kinetic energy in the small-strain limit.

5 Conservation Laws from Noether's Theorem

Theorem 5.1 (Noether conservation laws, Lean-proved)**.** *`Lean: QFT.NoetherTheorem`*
From the three symmetries of the RS voxel lattice $\mathbb{Z}^3$ (`VoxelSymmetry`):

Symmetry	*Conserved quantity*
Time translation ($t \to t + \varepsilon$)	*Energy* $H = T + V$
Space translation ($\mathbf{x} \to \mathbf{x} + \boldsymbol{\varepsilon}$)	*Momentum* $\mathbf{p} = m\dot{\mathbf{x}}$
Rotation ($\mathbf{x} \to R\mathbf{x}$)	*Angular momentum* $\mathbf{L} = \mathbf{x} \times \mathbf{p}$

6 Hamiltonian Mechanics

The Hamiltonian $H(\mathbf{q}, \mathbf{p}) = T + V$ gives Hamilton's equations:

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}. \tag{5}$$

In RS, these are the equations of motion for the recognition operator in phase space. The Poisson bracket $\{A, B\} = \partial_q A\, \partial_p B - \partial_p A\, \partial_q B$ becomes the commutator $[A, B]/(i\hbar)$ in the quantum limit.

7 Numerical Results

Result	RS Derivation	Classical Statement	Status
$F = ma$	Euler-Lagrange of J-cost action	Newton's 2nd law	✓
Inertia	Constant-p J-cost minimum	Newton's 1st law	✓
Action-reaction	Ledger double-entry	Newton's 3rd law	✓
Energy conservation	Time-translation symmetry	Conservation of H	✓
Momentum conservation	Space-translation symmetry	$\mathbf{p}$ conserved	✓
$\mathbf{L}$ conservation	Rotation symmetry	$\mathbf{L}$ conserved	✓

8 Lean Formalization Status

Claim	Lean theorem	Module	S
J-cost quadratic at $x = 1$	`Jcost_one_plus_eps_quadratic`	`Cost`	I
Noether theorem	`NoetherTheorem`	`QFT`	I
Ledger exactness (3rd law)	`ExactnessCert`	`Verification.Necessity`	I
Voxel symmetry (conservation)	`VoxelSymmetry`	`Verification.Necessity`	I
Hamiltonian emergence	`Foundation.RecognitionOperator`	`Foundation`	I
$F = ma$ from RS in Lean	—	—	I
Continuum limit of discrete RS	—	—	I

† **HYPOTHESIS**: Full Lean derivation of Newton's second law from the EL equations applied to the RS J-cost action. Falsifier: any known mechanical system violating $F = ma$ in the non-relativistic, non-quantum limit.

‡ **HYPOTHESIS**: Rigorous continuum limit $(\ell_0, \tau_0) \to 0$ of discrete RS dynamics gives continuous classical mechanics. Falsifier: observation of discrete tick signature in macroscopic mechanical systems.

9 Conclusion

Classical mechanics emerges from RS in the small-strain, non-relativistic limit. Newton's three laws, the principle of least action, Hamilton's equations, and the Noether conservation laws all follow from J-cost minimization ($J \approx \varepsilon^2/2$ at small strain). The derivation shows that Newtonian mechanics is not fundamental but the continuum limit of the discrete eight-tick recognition dynamics.

References

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